Proposition 13: a large integrality gap for the spectral distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Proposition14_SpectralLowerBound`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Proposition 13 (Section 3.4, p. 11, l. 389–390): *for every $r \geq 1$ there are graphs G_r, H_r on $n = \Theta(r^2)$ vertices such that $d_2(G_r, H_r) = \sqrt{r}$ and $\frac{1}{2\sqrt{2}} \leq d_2^{\mathrm{OT}}(G_r, H_r) \leq 1$.* The text after it (l. 391–393) concludes that there is no uniform bound $\delta_2 \leq C_\alpha(\delta_2^{\mathrm{OT}})^\alpha$ for $\alpha > \frac{3}{4}$. Theorem 3 (4) (p. 4, l. 127): $\delta_2(G_r, H_r) \geq \sqrt{r} \cdot \delta_2^{\mathrm{OT}}(G_r, H_r)$. Proof: App. F.1 (pp. 45–46), with the construction, Figure 7 and Lemma 49 ($\frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r) \leq 1$).

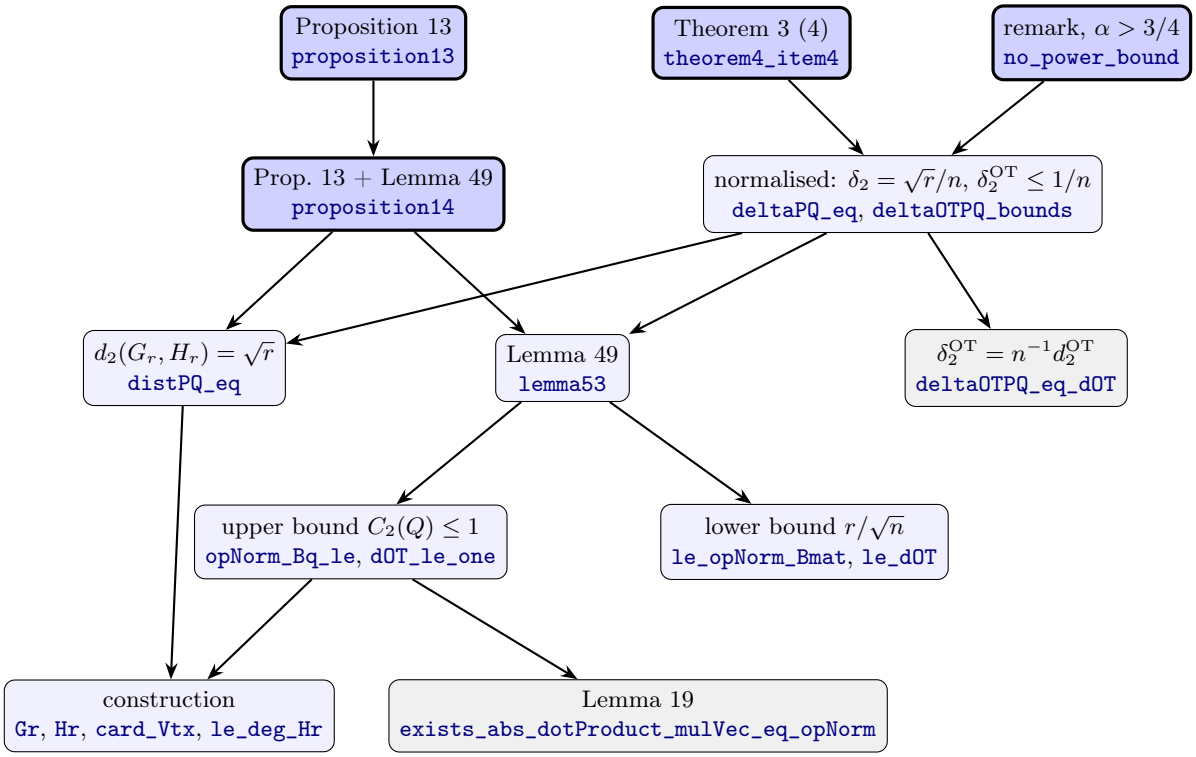
Lean. The graphs are `GsimOT.Prop14.Gr r`, `GsimOT.Prop14.Hr r` on `Vtx r`. For every $r \geq 1$:

- `proposition13`: $|V| = 4r^2 + 3r + 1$, $d_2(G_r, H_r) = \sqrt{r}$ and $\frac{1}{2\sqrt{2}} \leq d_2^{\mathrm{OT}}(G_r, H_r) \leq 1$ (exactly the statement);
- `proposition14`: Proposition 13 together with the bound $\frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r)$ of Lemma 49, i.e. the same with $\frac{1}{2\sqrt{2}} \leq \frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r)$; `lemma53`: Lemma 49;
- `theorem4_item4`: Theorem 3 (4);
- `no_power_bound`: for all $\alpha > \frac{3}{4}$ and $C \in \mathbb{R}$ there is $r \geq 1$ with $C\,\delta_2^{\mathrm{OT}}(G_r, H_r)^\alpha < \delta_2(G_r, H_r)$.

2 Where the proof lives

File	Content
`Construction.lean`	the vertex set $X \cup M$ (`Core`, `Marker`, `Vtx`), the graphs (`graph`, `Gr`, `Hr`), $\lvert V\rvert = 4r^2 + 3r + 1$ (`card_Vtx`), degrees (`le_deg_Hr`).
`SpectralDistance.lean`	$d_2(G_r, H_r) = \sqrt{r}$ (`distPQ_eq`) and elementary facts on ℓ^2 and spectral norms.
`Lemma53.lean`	Lemma 49: upper bound via $Q = \frac{1}{2r}\sum_i P_{\sigma_i}$ (`dOT_le_one`), lower bound via the test vector $\mathbf{1}_R$ (`le_dOT_of_deg`, `le_dOT`).
`Proposition14.lean`	Proposition 13 (`proposition13`, `proposition14`), Theorem 3 (4), and the remark (`no_power_bound`).
`Common/OTDistances.lean`, `Common/OperatorNorm.lean`	$B_{2\to 2}$, d_2^{OT}, $\delta_2^{\mathrm{OT}} = n^{-1} d_2^{\mathrm{OT}}$ (`deltaOTPQ_eq_dOT`), Lemma 19.

3 Proof layout



Arrows mean "uses". Blue: this folder; grey: `GsimOT/Common`. Nothing is cited.

4 The proof step by step

Construction (p. 45, l. 1399–1411; Figure 7, p. 45). Core $X = \{u\} \cup \{v_{i,k} \mid i \in [2r], 0 \le k \le r\}$ is `Core r` = `Option (Fin (2r) × Fin (r+1))` (u = `none`); markers $M = C \cup T_1 \cup \cdots \cup T_{2r}$ are `Marker r` (clocks `inl j` $= c_{j+1}$, tags `inr (i,t)`). $N(v_{i,k}) = T_i \cup \{c_1, \ldots, c_k\}$ in both graphs, u is isolated in G_r and adjacent to all clocks in H_r: one definition `graph` with a Boolean flag. $|V| = 4r^2 + 3r + 1$ is `card_Vtx`; H_r has minimum degree $\ge r$ (`le_deg_Hr`).

$d_2(G_r, H_r) = \sqrt{r}$ (p. 45, l. 1412–1416). Lower bound: the spectral norm is at least the Euclidean norm of a row (`sqrt_row_le_opNorm22`), and the row of $\pi(u)$ has $\ge r$ entries ± 1 (`sqrt_le_opNorm_alignedDiff`). Upper bound: for the identity, $A_G - A_H$ is minus a star with r leaves, of norm $\sqrt{r}$ (`opNorm_diff_refl_le`). Together: `distPQ_eq`.

Lemma 49, upper bound (pp. 45–46, l. 1418–1436). σ_i shifts branch i one step along $u \to v_{i,0} \to \cdots \to v_{i,r} \to u$ (`shift`, `tau`); $Q = \frac{1}{2r} \sum_i P_{\sigma_i}$ fixes the markers (`Qs`, `Qs_isDS`). The vectors $d_{xy} = \mathbf{1}_{N_G(x)} - \mathbf{1}_{N_H(y)}$ (`dv`) are $-\mathbf{1}_{T_i}$, $-\mathbf{1}_{\{c_{k+1}\}}$, $\mathbf{1}_{T_i}$ on the support. $B_{2\to 2}(G_r, H_r, Q)$ has the block form $\left(\begin{smallmatrix} 0 & K \\ K^\top & 0 \end{smallmatrix}\right)$ (`Bq_mulVec_cc`, `Bq_mulVec_mm`, `dotProduct_Bq`). The paper computes $\|K^\top K\|_2 = \|\frac{1}{r} \sum_i \mathbf{1}_{T_i} \mathbf{1}_{T_i}^\top + \sum_j \mathbf{1}_{c_j} \mathbf{1}_{c_j}^\top\|_2 = 1$. The Lean proves the corresponding quadratic-form bound $\zeta^\top K^\top K \zeta = \sum_{x,y} Q(x,y)(d_{xy} \cdot \zeta)^2 \le \|\zeta\|^2$ (`sum_branch_le`, `sum_qc_dvDot_le`: Cauchy–Schwarz on each tag block, orthogonality of the clocks), and concludes $|y^\top B x| \le 1$ for unit x, y via Lemma 19 (`opNorm_Bq_le`), hence $d_2^{\mathrm{OT}} \le 1$ (`dOT_le_one`).

Lemma 49, lower bound (p. 46, l. 1437–1446). For every $R \in \mathrm{DS}$, the unit vector $\mathbf{1}_R(x,y) = \sqrt{R(x,y)/n}$ satisfies $(B\mathbf{1}_R)(x,y) = \mathbf{1}_R(x,y)(\deg_G x - \deg_H y)$, and the rows of u give $\|B\mathbf{1}_R\| \ge r/\sqrt{n}$. This is proved for arbitrary graphs and any vertex u with $|\deg_G u - \deg_H y| \ge c$ for all y (`le_opNorm_Bmat`, `le_dOT_of_deg`), then specialised (`le_dOT`).

Proof of Proposition 13 (p. 46, l. 1447–1448). Lemma 49, $d_2 = \sqrt{r}$ and $\frac{r}{\sqrt{n}} \ge \frac{1}{2\sqrt{2}}$ since $8r^2 \ge n$: `proposition14`; `proposition13` drops the middle term.

Theorem 3 (4) and the remark (p. 4, l. 127; p. 11, l. 391–393). $\delta_2(G_r, H_r) = \sqrt{r}/n$ (`deltaPQ_eq`) and $0 \le \delta_2^{\mathrm{OT}}(G_r, H_r) \le 1/n$ (`deltaOTPQ_bounds`, via `deltaOTPQ_eq_dOT`) give `theorem4_item4`. For `no_power_bound`: for $\alpha > \frac{3}{4}$ choose $a = \min\{\alpha, 1\}$ and r large; $(\delta_2^{\mathrm{OT}})^\alpha \le n^{-a}$ and $n \le 8r^2$ give $C(\delta_2^{\mathrm{OT}})^\alpha \le C8^{1-a}r^{2(1-a)}/n < \sqrt{r}/n$. Only the upper bound $\delta_2^{\mathrm{OT}} = O(n^{-1})$ is needed, not $\Theta(n^{-1})$.

5 Assumptions and trusted base

No cited results; the hypothesis is $r \ge 1$. Axioms: `propext`, `Classical.choice`, `Quot.sound`.

6 Notes on the formalisation

- The upper bound of Lemma 49 is proved through the quadratic form of $K^\top K$, and only $C_2(Q) \le 1$ is proved, which is what $d_2^{\mathrm{OT}}(G_r, H_r) \le 1$ requires (the paper computes $C_2(Q) = 1$).
- The lower bound of Lemma 49 is proved for arbitrary graphs.
- `proposition14` records the intermediate bound $\frac{r}{\sqrt{n}} \le d_2^{\mathrm{OT}}(G_r, H_r)$ used in the proof of Proposition 13; `proposition13` is derived from it.

7 How to check this result

```
lake build GsimOT.Proposition14_SpectralLowerBound.Proposition14
lake env lean scripts/CheckAxioms.lean
```

The table in `Proposition14_correspondence.pdf` lists every step with file and line.